

Module 05 Lab: Contribution Journal

Course: ITAI 1371

Date: September 19, 2026

Min:

For this lab, my first contribution was troubleshooting the notebook since it started out completely unrunnable, so I dived in and resolved the formatting issues to make it readable and executable for the group. Once the notebook was stable, I assigned each teammates including myself to roles so we could effectively work on it. My roles are in part 1 of the data acquisition and configuration stage of the process to create our baseline data. Also, I was responsible for the analytical evaluations of Q2 and Q3 periods by analyzing the mathematics of One-Hot Encoding and explaining the reason Feature Scaling has no effect on Decision Tree operations.

Camille:

My contribution focused on Task 1, where I worked with the 177 missing values in the Age column. At first, I misunderstood the 177 as the value that needed to be entered, but I learned that it represented the total number of missing ages. I then calculated the median age and used `.fillna()` to replace the missing values correctly. After running the code, I verified that there were no remaining missing values in the Age column. Working through my initial mistake helped me better understand the code and the purpose of median imputation.

Chris:

For Lab 05, I worked on Task 3 for this lab where we used “StandardScaler” from scikit-learn to scale “Age” and “Fare” features in the Titanic data set. The code created a StandardScaler, selected “Age” and “Fare” features, fit the scaler to the data and then used the transformed data. The output of the scaled data was then checked to ensure it had been properly scaled. To test this section out independently, I first completed Task 2 by using one-hot encoding on “Sex” and “Embarked” features of the data, thus creating the “encoded_df” DataFrame that Task 3 needed. After that, I checked that “Age” and “Fare” features were properly converted to scaled features. I then cleaned up the completed solution for Task 3 and added it to the group’s notebook for Lab

05. The rest of the lab was completed by other group members and they did the final submission for the lab.

Samantha:

My contribution to this lab was completing Question 1 of the Knowledge Check and helping with the final preparation of our group assignment. For my question, I explained why the median can be a better option than the mean when filling in missing values in a dataset. I learned that the median is less affected by outliers or unusually high and low values, which can make it a better representation of the typical data. I also helped review the notebook after my group members completed their assigned sections. I checked that the different tasks were completed, ensure the notebook was organized, and helped prepare everything for the final submission. Working on this lab gave me a better understanding of data preparation and showed me why cleaning and preparation are important before using it for machine learning.

Rigoberto:

For my part of the group project, I worked on **Task 2: One-Hot Encoding**. I converted the categorical `Sex` and `Embarked` columns into numerical columns using one-hot encoding. I used `pd.get_dummies()` with `drop_first=True` to avoid creating redundant columns. The resulting columns included `Sex_male`, `Embarked_Q`, and `Embarked_S`. I also displayed the first few rows to verify that the encoding worked correctly. This encoded data can then be used for later steps such as scaling and building the machine learning model.